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Chapter 1

Advanced Customer Scheduling

Overview



Introduction

The Advanced Customer Scheduling module aims to cater for the special requirements of repetitive supply chain industries, for example in the automotive or white goods sectors.

The application allows you to manage and monitor complex supply contracts and automates the receipt and validation of scheduled data enabling you to intervene on an 'exemption only' basis.

Advanced Customer Scheduling provides the following functions:

- Master File Maintenance
- Schedule manipulation for entering/maintaining schedule data
- Facilities to cater for different types of schedules:
 - Releases
 - DCIs
 - RANs
- Flexible Comparison Processes
- Multiple Authorisation Periods
- Reconciled Requirements
- Maintenance of Contract Cumulatives
- Contract Pricing (including a zero pricing facility).
- Sleeper processes to receive, validate, compare, and accept schedule data

Master File Maintenance

The main file maintenance functions are:

- Company
- Customer
- Contract

These are repetitive functions which filter the default values down from company to customer, and from customer to contract level. This minimises the amount of keying involved in setting up a new contract whilst allowing the flexibility of overrides at the lower levels.

A contract can have two sets of parameters: one which determines how the system should handle 'Release' type schedules; and secondly a second set of parameters which determine how the system should process DCI or RAN type schedules.

Advanced Customer Scheduling also provides maintenance facilities for Contract Pricing and Comparison Rules.

The Comparison Process

You can specify comparison rules for monitoring variances between subsequent versions of schedules against a contract.

Comparison rules can have five different types of rule element attached to them, with corresponding date ranges. The combination of element/date range is unique on the rule. The elements determine where the factor which is being compared should be measured, for example the quantity within the date range, or the quantity at the date range end.

Each combination of element/date range on a rule can have tolerance values (positive and/or negative) for: the quantity difference; the percentage difference; and the sales value difference between subsequent schedules.

Each rule element/date range set of tolerances has a corresponding severity level between 0 and 9. A variance which is outside the tolerance with a severity of 4 or over will cause the new schedule data to fail comparison and prevent acceptance. A variance of severity less than 4 will record the failure as an exception without preventing acceptance.

Rules may then be attached to Contract Release and /or contract DCI defaults via Contract Maintenance. Rules are attached to contracts in combination with comparison methods, which tell the system which previous version of a contract's schedule to use as a base for comparing the new data.

Reconciled Requirements

When new schedule data is accepted into the current schedule files, it goes through a process which creates reconciled requirements. In brief, this is the process of converting the raw data, as sent by the customer, into requirements to be fed into Advanced Shipping.

This involves adjusting the data by the following factors (as applicable), using the reconciliation and arrears processing methods linked to the contract:

- Goods in transit.
- Arrears from previous schedules
- Ship lead time adjusting (converting required by dates into ship by dates).
- Delivery frequency consolidation. This is consolidating deliveries according to customer and stock room calendars.
 - when do they require the goods?
 - what days can we ship to them?
 - what days can they receive shipments?
- Pack rounding. Contracts may be set up to perform pack rounding with an associated pack rounding percentage. This will round the shipment quantity up or down to the nearest pack size on the basis of the pack rounding percentage.
- After reconciling a new schedule, any shortfall or over resulting from pack rounding is held against the contract for reconciling with the next version of the schedule.



Note: You may also explode a single 'raw' requirement into multiple 'reconciled' requirements where the customer sends, for example, monthly requirements.



Note: RANs are only ship lead time adjusted, they are never pack rounded.

Contract Cumulatives

The system stores and monitors the following data:

- Cumulative despatches against the contract.
- High water cumulatives. These are the highest quantity of requirements by authorisation period that the customer has ever committed to. Particularly important for firm and material authorisation periods where there may be a contractual obligation against the customer to pay for the high water values.
- Engineering change levels. The system maintains a log of valid engineering change revision levels against the contract, with valid dates/quantities. The current engineering change level is held on the contract and there is a facility for 'holding' new schedules with a change of engineering levels.

Contract Pricing

You can choose between:

- Contract pricing, using effective and charge from dates, with a retrospective invoicing report facility.
- Normal pricing which will price despatches out of Advanced Shipping.

In addition, contract pricing will permit zero pricing for samples or free of charge shipping during pricing negotiations.



Note: The pricing is established at company level.



Note: If you have installed Advanced Pricing only, contract pricing can be installed for Advanced Customer Scheduling.

Schedule Manipulation

You can review and/or maintain new schedule data and send the resulting schedule through the appropriate sleeper processes.

Schedule manipulation may be used to manually enter schedule data which has arrived by, for example, fax. Alternatively, existing (current) schedule data may be maintained. In this case the system will copy the data to the new files to allow manipulation. The manipulated data may then be sent back through the sleepers to update the current schedule data.

You can also review comparison exceptions, i.e. see which comparison rule tolerances have been exceeded and with what severity. It will also allow you to rerun the comparison process on-line, following manipulation.

You can also produce reconciled requirements on-line and review them prior to acceptance.

Schedule Maintenance caters for new schedule data which may still be in the Application Interface (EDI) files, in addition to catering for data in the 'current' files. Hence schedules which have failed the EDI receive and validate sleeper may be reviewed and maintained.

Error Messages

Error messages are reported extensively during schedule manipulation. In many instances it will be possible to either amend the schedule data, or change some of the contract values to correct an error.

All error messages that arise from the customer schedule validation, comparison and acceptance processes have a default severity assigned to them. These default severities can be overridden using the Maintain Message Severities facility on the Utilities menu.

If the actual severity is less than 4, the error message is treated as a warning only. If it is greater than or equal to 4 then it is treated as a normal error message, and you will not be able to accept changes to the screen until such errors are corrected.

Error Correction

This section details the error conditions that are currently trapped by the acceptance process, and describes how schedule manipulation will be used to correct the error.

Error	Option	Action
Unable to match to an active contract.	Process EDI Error Transactions	Select the appropriate schedule, and use the contract match facility. Possibly enter a value in the link field.
Contract already has a new schedule, of this type.	Process EDI Error Transactions	Delete the schedule that is already in the new files, or take the appropriate action to allow it to

Error	Option	Action
		progress to 'current'.
Duplicate release.	Process EDI Error Transactions	Delete the duplicate.
Schedule amendment - previous issue details do not match.	Process EDI Error Transactions	Change the previous document information to match the current schedule, or leave this schedule until the version to which it relates does become current.
Schedule out of sequence.	Process EDI Error Transactions	Delete the schedule, since a later version has already become current.
Invalid effectivity dates.	Contracts Maintenance Process EDI Error Transactions	The invalid dates can be changed, or the contract duration can be extended to cover the specified period.
Manual review.	Process New Schedules Process EDI Error Transactions Schedule Maintenance	You will confirm that the schedule can progress to the next stage in the process.
Schedule amendment (RAN), the original RAN does not exist or has been completed.	Process New Schedules	Delete the RAN amendment and reconcile situation with customer, then update the schedule requirement.
Date validation.	Process New Schedules	Correct the date in error.
Time Validation.	Process New Schedules	Correct the time in error.
Date of new release.	Process New Schedules	Correct the date.
Unknown despatch note number.	Process New Schedules	Correct the customer quoted despatch note numbers. A list of despatches can be displayed.

Error	Option	Action
Invalid final delivery address code and additional destination, consignee, freight forwarder, or carrier.	Process New Schedules	Change Schedule Header Additional Details Screen.
Invalid authorisation period end dates.	Process New Schedules	Correct the date.
Invalid authorisation period cumulatives.	Process New Schedules	Correct the cumulatives.
Invalid article status code.	Process New Schedules	Change Schedule Header Additional Details Screen.
Design revision number.	Process New Schedules	Change Schedule Header Maintenance Screen.
Invalid engineering change number.	Process New Schedules	Change Schedule Header Maintenance Screen.
Invalid first date display format.	Process New Schedules	Change using Release Details Maintenance - More Keys Screen.
Invalid authorisation status.	Process New Schedules	Is only checked, and will only need correction, if the authorisation method on the contract is '01'. If correction is required, this can be amended using Release Details Maintenance - More Keys Screen.
Invalid call off reference.	Process New Schedules	Change the call off reference using Release Details Maintenance - More Keys Screen.
Any comparison errors.	Process EDI Error Transactions Schedule Maintenance	Correct by manipulating the schedule header in conjunction with the schedule detail requirement.

Sleeper Processes

There are three sleeper processes which are run as background tasks.

Validation Sleeper

The first sleeper is the validation sleeper. The data queue which feeds this program contains EDI records, and/or manual schedule manipulation records.

The sleeper receives data in its raw state, and performs fundamental validation (such as identifying the contract). Where it can, data from the Application Interface (EDI) files are transferred to the new Advanced Customer Scheduling data files.

Comparison Sleeper

The second sleeper is the comparison sleeper. This takes the new Advanced Customer Scheduling data and compares it to previous versions of schedule data for the same contract and schedule type.

This is performing more critical validation as specified by the comparison rules, methods and elements specified on the contract. Its primary function is to highlight inconsistencies.

An example of this type of check is frontloading. This is where the customer tries to shift an unforeseen increase in demand up the supply chain. As authorisation periods are constantly moving and data is generally stated as cumulatives, it could be quite difficult to spot such circumstances manually.

Acceptance Sleeper

The third sleeper process is the acceptance sleeper. This takes scheduled data which has passed the first two sleeper processes and updates the Advanced Customer Scheduling current schedule data.



Note: The current data will contain only the schedule details that are effective in the system. All previous schedules will be kept in the History files until purged by the relevant option.



Note: This means that for a contract operating releases and DCIs, there will be two sets of data in the current data files.

As new sets of records come through the first sleeper, they update the new schedule data files. The acceptance sleeper writes the current set of schedule records to the history files, then moves the new records into the current files.

The acceptance sleeper also takes the newly accepted raw requirements and creates reconciled shipping requirements out of them.